

Power-Efficient W-Band (92–98 GHz) Phased-Array Transmit and Receive Elements With Quadrature-Hybrid-Based Passive Phase Interpolator

Sadia Afroz, *Student Member, IEEE*, Hyunchul Kim, *Student Member, IEEE*,
and Kwang-Jin Koh¹, *Member, IEEE*

Abstract—This paper presents a power-efficient phased-array transmit and receive (T/R) channels adopting phase shifters (PSs) utilizing quadrature-hybrid-based power-domain phase interpolator. The W-band T/R channels are implemented in GlobalFoundries 0.13- μm SiGe BiCMOS process (GF8XP) for large-scale phased-array applications. At 94 GHz, the proposed Rx channel achieves 137-dB-Hz dynamic range (DR) with 18-mW dc power consumption (P_{diss}), one of the smallest P_{diss} at W-band. The measured Rx channel gain is 17–23 dB with 5.8–6.6-dB noise figure (NF) over 92–98 GHz. The rms gain error and rms phase error in the Rx channel are less than 2 dB and 6° , respectively, at 92–100 GHz. The input $P_{-1\text{dB}}$ is -31 dBm at 94 GHz. The Tx channel produces 7.4-dBm saturated output power with 49.5-mW P_{diss} at 94 GHz resulting in 11% of collector efficiency, one of the highest Tx channel efficiencies reported so far. The P_{diss} at quiescent bias point of the Tx channel is 26 mW. The measured Tx channel gain is 7.5–12 dB and the rms gain error is <1.28 dB over 85–105 GHz. The measured rms phase error in the Tx channel is $<1.5^\circ$ at 94 GHz and remains lower than 8° when driving the Tx channel up to near the power saturation. The peak collector efficiency and PAE of the integrated class-AB power amplifier is 23.5% and 18.2%, respectively, at 94 GHz, one of the highest efficiency performances at W-band. The chip sizes of the Rx and Tx channels excluding I/O pads are $1.2 \times 0.55 \text{ mm}^2$ and $1.4 \times 0.55 \text{ mm}^2$, respectively.

Index Terms—BiCMOS, hybrid coupler, phase shifter (PS), phased array, quadrature coupler, vector modulator (VM), W-band.

I. INTRODUCTION

THANKS to a small wavelength combined with a low loss free-space transmission window around 94 GHz, the W-band (75–110 GHz) provides a promising opportunity for highly directive wireless communications and high-resolution

Manuscript received July 19, 2017; revised November 24, 2017 and January 22, 2018; accepted February 15, 2018. Date of publication April 3, 2018; date of current version May 24, 2018. This paper was approved by Associate Editor Hossein Hashemi. This work was supported by DARPA Strategic Technology Office through the program “Low-Power 94-GHz T/R Element Design for Multifunctional Arrays in Advanced Silicon Technologies” under Contract HR0011-14-C-0128. (Corresponding author: Kwang-Jin Koh.)

S. Afroz is with Qualcomm Inc., San Diego, CA 92121 USA (e-mail: lisa027@vt.edu).

H. Kim and K.-J. Koh are with the Electrical and Computer Engineering Department, Virginia Tech, Blacksburg, VA 24061 USA (e-mail: hckim@vt.edu; kkoh@vt.edu).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JSSC.2018.2810178

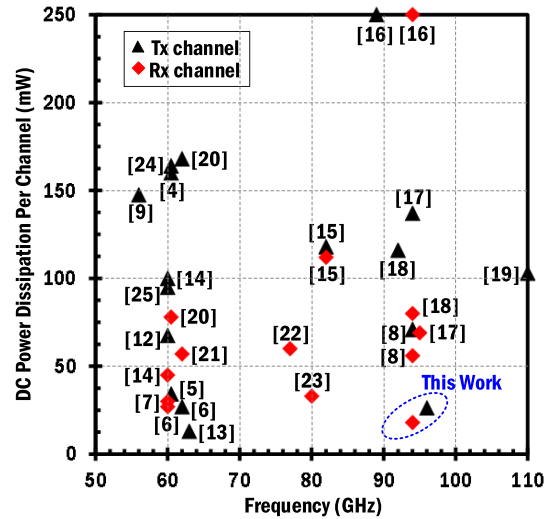


Fig. 1. DC power dissipation per array element of phased-array front ends at millimeter-wave frequencies.

imaging systems [1], [2]. Recently, the 94-GHz band is also gaining attraction for highly sensitive defense radar sensors with less than several degrees of beamwidth [3]. The high directivity requires a large antenna aperture, necessitating a large phased array. When the array size increases, the required output power per array element may decrease, allowing the design of phased-array electronics using a low breakdown but fast silicon process. This heralds opportunity for the silicon technology as a system integration platform for large array constructions. However, for the large silicon array to be of practical use, it is paramount to maintain a high power efficiency (η) in the array element.

Fig. 1 summarizes dc power dissipation (P_{diss}) per phased-array transmit and receive (T/R) elements of recent state-of-the-art (SOA) silicon-based phased arrays. Although significant efforts have been made to improve η at 60 GHz, the P_{diss} per element at W-band is still high. The high P_{diss} necessitates huge heat sinks around the array sensors when a large number of elements (typically, >1000 elements) are assembled, increasing the size and the weight, and thus confining the field mobility of the array sensors. In the target large array assumed in this paper, the required element

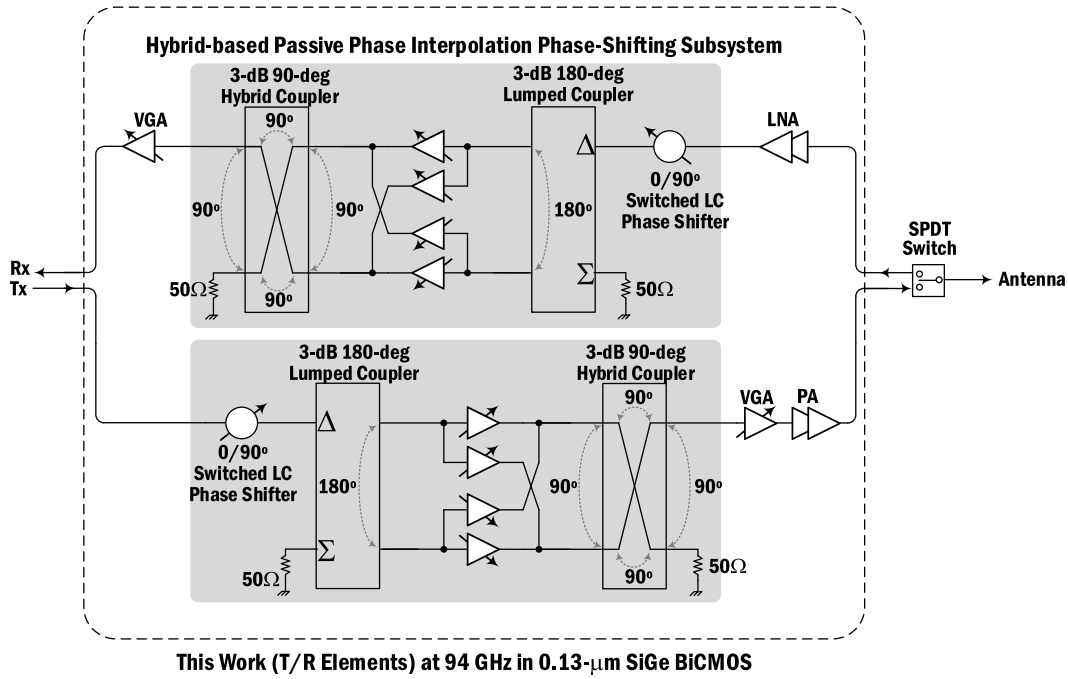


Fig. 2. Block diagrams of the phased-array T/R channels utilizing the quadrature hybrid-based passive phase interpolation phase-shifting subsystem.

output power (P_{out}) is typically 3–5 mW [3]. In such a modest P_{out} , the loss (P_{loss}) of passive phase shifters (PSs) becomes a major culprit for degrading η . To improve phased-array performance, different types of phase shifting schemes have been implemented in the baseband [4]–[7], the local oscillator (LO) [8]–[11], or the RF domains [12]–[25].

In W -band, the RF phase shifting architecture is preferable to the baseband or LO phase shifting schemes for the hardware simplicity by avoiding complex frequency conversion units [9]–[11]. Potentially, this can lead to a high power efficiency with a compact area. However, the benchmarking of P_{loss} of the RF PSs based on switchable LC passive networks using similar SiGe BiCMOS processes reveals that the P_{loss} could be as high as 17–20 dB with 4-bit phase resolution [23], claiming substantial dc power for the loss compensation. This high loss is mainly originated from the switch loss implemented in available 0.13- μm NMOS transistor in the SiGe technologies and could be improved with more scaled and faster CMOS processes. The fundamental limitation of the switched LC network-based PSs, however, comes from the tradeoff between loss and phase resolutions, namely, increasing phase resolution needs cascading more switched LC sections, hence causing more loss from the cascaded passive networks.

The passive reflection-type PS (RTPS) utilizes a 90° hybrid coupler with varactor-tuned LC loads to create 180° of phase span and typical loss is >7 dB at W -band [18]. However, additional active or passive balun and switch network are necessary to expand the phase range to 360°, incurring an extra loss and power dissipation and thus making it less attractive for power-efficient PS systems. The conventional vector modulator (VM) type active PS can achieve a high phase resolution by interpolating quadrature phases in a continuous way but at

the cost of significant dc power consumption with limited gain and linearity performance at W -band [15]–[17].

This paper addresses the power efficiency issue at W -band (92–98 GHz) phased-array T/R channels integrated in 0.13- μm SiGe BiCMOS process. Afroz and Koh [26] present a distributed phase shifting subsystem relying on the 90° hybrid coupler-based phase interpolator. The main technical advancement in this paper is the renovation of the distributed phasing system by eliminating loss-ridden single-pole double-throw (SPDT) switch networks, achieving some of the best power efficiencies reported so far at W -band and beyond, as shown in Fig. 1. More details on the phase shifting system architecture are described in Section II. The specific building block designs and experimental results are elaborated in Sections III and IV, respectively. Finally, this paper is concluded in Section V.

II. PHASE SHIFTING SUBSYSTEM ARCHITECTURE

Fig. 2 shows complete phased-array T/R channel architectures based on the proposed phasing subsystem. Fig. 3 illustrates details on the PS architectural improvement compared with the prior implementation in [26] and its phasing operations. Fundamentally, the proposed phasing technique is relying on the phase interpolation process, which requires differential quadrature basis signal sets to synthesize full 360° of different phase states. The prior work in [26] generates the basis signals in a distributed manner by a series of 90° and 180° hybrid couplers and by inserting an SPDT switch in between the hybrids to preselect one of the quadrature phases as shown in Fig. 3(a) (top). The main issue here is the power loss—theoretically, the 90° hybrid claims 3-dB loss since one of its output is terminated with 50 Ω and the SPDT loss would be 2–2.5 dB at W -band, practically totaling 6–6.5 dB of P_{loss}

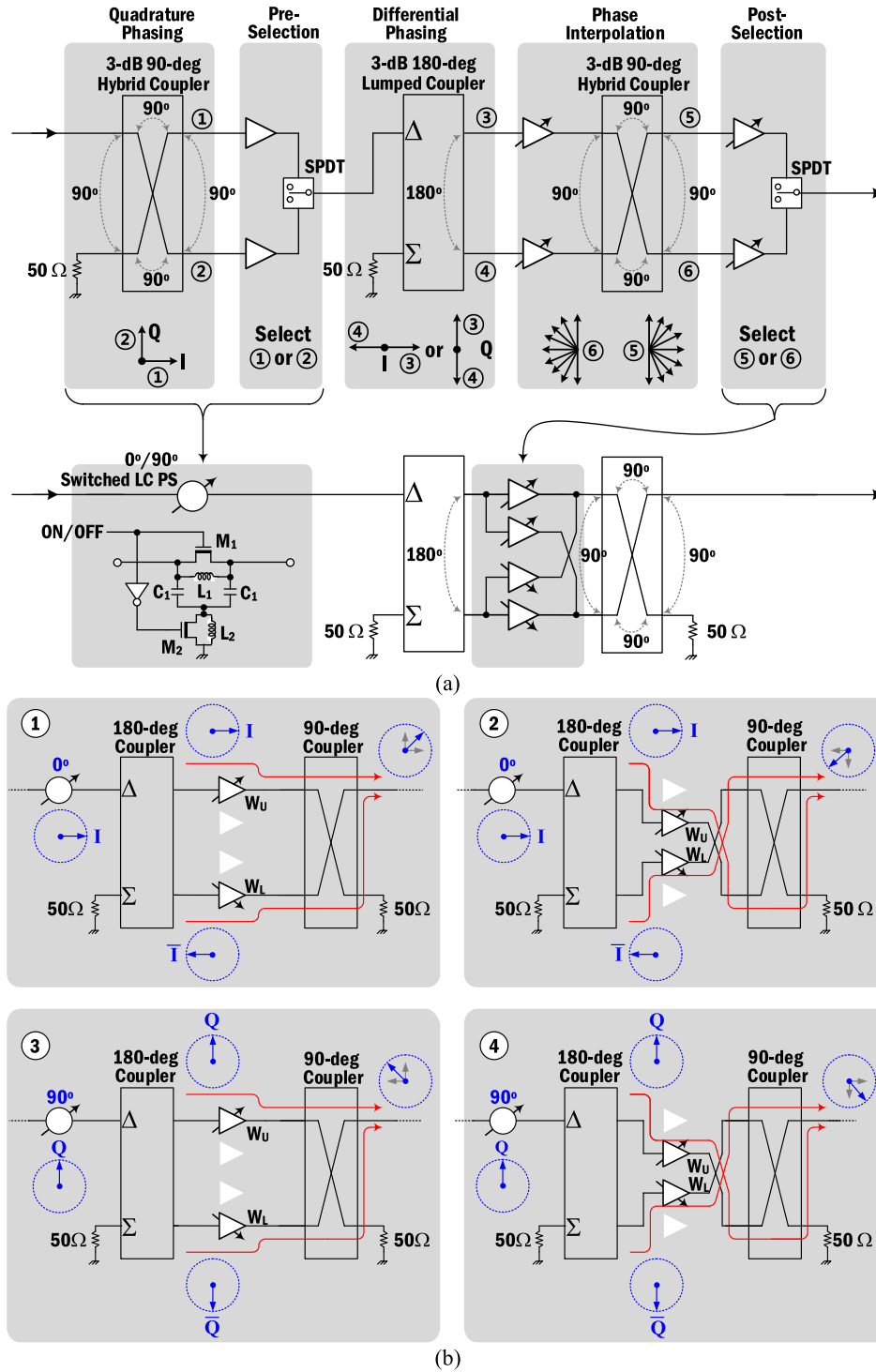


Fig. 3. Block diagrams of the quadrature hybrid-based passive phase interpolation phase-shifting subsystem. (a) System modification of the quadrature hybrid-based phasing subsystem for low loss and area compactness. (b) Four quadrant phase synthesis. ①: first quadrant phase. ②: second quadrant phase. ③: third quadrant phase. ④: fourth quadrant phase.

when considering about 1 dB of ohmic loss in the 90° hybrid at W-band.

To compensate the loss, an amplifier is cascaded in between the 90° coupler and SPDT in Fig. 3(a), incurring dc power dissipation. In the newly proposed system as shown in Fig. 3(a) (bottom), this hybrid amplifier SPDT chain is replaced by a switched LC network-based 0°/90° switched

LC PS [23]. This not only reduces the loss to 3–3.5 dB in the same 0.13- μm SiGe BiCMOS process but also shrinks the area factor significantly for a complete T/R module to be integrated into $\lambda/2 \times \lambda/2$ antenna array lattice. Note that the main source of the loss in the switched LC PS is the finite ON/OFF resistance of the NMOS switch, which can be improved substantially with more scaled CMOS process.

To further reduce the loss and the area factors, the post phase selection stage, comprised of a variable gain amplifier (VGA) and another SPDT, is replaced by a quad VGA cell and the phase selection function is implemented with switching the quad VGAs. This is illustrated in Fig. 3(b). At 0° state in the front-end LC PS and by switching the outer VGA pair to inner VGAs, the first quadrant phases $\{=\tan^{-1}(W_L/W_U)\}$ and their differential pairs in the third quadrant phase domain can be created [see Fig. 3(b) ① and ②]. Essentially, the inherent functions of orthogonal phase splitting and power combining in the 90° coupler create a complex weight, $jW_U + W_L$ or $-W_U - jW_L$, for a given differential input, enabling the 90° coupler to play the role of a phase interpolator. Similarly, if we select the quadrature component of the input signal generated from the $0^\circ/90^\circ$ switched LC PS, the second and the fourth quadrant phases can be synthesized by manipulating the signal flow by the quad VGAs [see Fig. 3(b) ③ and ④], completing a 360° phase rotation.

To account for loss factors, the lumped 180° coupler causes 0.5–1-dB ohmic loss at 94 GHz. During the phase interpolation by the 90° coupler, a half power is split to $50\text{-}\Omega$ termination, thus inflicting a 3-dB loss in addition to 0.5–1-dB ohmic loss. Therefore, when including 3–3.5-dB loss from the $0^\circ/90^\circ$ passive LC PS, the total PS loss could be 7–8.5 dB which can be compensated by the weighting VGAs with a relatively small dc power penalty. Compared with the conventional active current-mode vector summing PSs, the power-mode vector summing in the hybrid is a linear passive process and therefore can result in a low phase distortion at a high input power level. The linearity of the PS is mostly dependent on the VGAs and can be much better than conventional current-combining active VMs.

Fig. 4 illustrates detailed linearity comparison between the current-mode VM and power-mode VM. C_o and r_o represent the nonlinear output capacitance and nonlinear output resistance of transconductor (G_m) in the current-mode VM or VGA in the power-mode VM. In the conventional three-stack or two-stack Gilbert cell-based VM, the variable gain G_m cells are loaded by each other's nonlinear output passive components. Whereas in the proposed power-mode VM in Fig. 4(b), each VGA output is isolated by the 90° hybrid coupler. Namely, ports ① and ④ are isolated due to the quadrature phasing in each hybrid arm and the signals are added at the pure passive hybrid output port. Because of the output ports isolation, the nonlinear loading effect becomes lower in the proposed power-mode VM, resulting in better linearity performance compared to the Gilbert cell-based current-mode VM.

The details on the linearity simulation result are shown in Fig. 4(c) for 45° phase state. For a fair comparison, the overall P_{diss} are same with 2.2-V supply voltage for all cases (all NPN heterojunction bipolar transistor (HBT) has $l_e = 3\text{ }\mu\text{m}$) and all ports are matched with $50\text{-}\Omega$ with ideal passive components. Thus, the net circuit nonlinearity comes from the transistor nonlinearity. Between the three-stack and the two-stack Gilbert cell topologies, the large parasitic capacitance in the three-stack quad-switching unit leads to a high loss, resulting in about 12-dB lower gain in the three-stack topology compared with the two-stack Gilbert cell.

For better voltage headroom with less loss, the two-stack architecture would be a better choice for high-frequency applications [15], [17]. It should be noted that because of the full I/Q differential operation, the input transistors of the Gilbert cell-based VM experience 3-dB lower input power. The simulated $\text{IP}_{-1\text{dB}}$ is 3 dB higher in the proposed VM at 94 GHz, resulting in -9 dBm of $\text{IP}_{-1\text{dB}}$ with the same P_{diss} as in the Gilbert cell-based current-mode VMs, although the individual input transistors of the power-mode VM undergo 6-dB higher input power (or $2\times$ higher input voltage swing) compared with the Gilbert cell-based VM [see Fig. 4(c)]. It is worthwhile to mention that if the current density (J_c) of all RF input transconductors are set to same (e.g., 1 mA bias current for each VGA input transistor while keeping 2 mA tail current for each differential pair in the Gilbert cell VMs), the P_{diss} of the power-mode VM will be half of the Gilbert cell-based VM. This decreases the gain of the power-mode VM by 2.5 dB but $\text{IP}_{-1\text{dB}}$ increases to -5 dBm as shown in Fig. 4(c), still 4.5 and 15.5 dB of $\text{OP}_{-1\text{dB}}$ improvements over the $\text{OP}_{-1\text{dB}}$ of the two-stack and the three-stack Gilbert cell-based VMs, respectively, albeit with 50% P_{diss} reduction. The transmission line (T -line)-based quadrature hybrid can be wideband and the phase response of the hybrid is robust against impedance variations at different VGA gain states, resulting in a low phase error for a wide frequency range at W -band.

III. BUILDING BLOCKS DESIGN AND CHARACTERIZATIONS

The T/R channels are designed in GlobalFoundries $0.13\text{-}\mu\text{m}$ SiGe BiCMOS technology (GF8XP, $f_T/f_{\text{max}} = 270/330\text{ GHz}$). The process has seven metal layers with the top two being thick aluminum metal layers (AM and LY layers). All inductors are designed with the AM layer and a patterned ground (GND) plane using M1/M2 layers. To minimize parasitic inductance in the supply (V_{CC}) and GND planes while maximizing ac decoupling through the V_{CC} and the GND planes, multiple wide-metal V_{CC} and GND planes are alternatively stacked and connected together with vias. In order to prevent potential low-frequency oscillation, multiple low Q -capacitor banks are also inserted between the V_{CC} and the GND planes.

A. LNA and VGAs

In the receive channel shown in Fig. 2, a two-stage cascode LNA proceeds the phasing circuitry to provide sufficient gain to reduce the noise effect from the following PS. The schematic, shown in Fig. 5(a), serves as a common fundamental building block for LNA and VGAs in this paper. Thus, it is carefully designed and optimized to achieve a high gain efficiency. The peak f_T of the NPN HBT degrades down to 240 GHz from 270 GHz after including electromagnetic (EM) wave model of the transistor layout interconnects, while there is no noticeable change in the NF_{min} (see Fig. 6). The bias current density of the transistors is chosen to $J_c = 0.67\text{ mA}/\mu\text{m}$ for a minimum NF with a low dc power penalty at 94 GHz. At this current density, the f_T including transistor interconnects is $\sim 200\text{ GHz}$. All transistors are sized to have the emitter length of $3\text{ }\mu\text{m}$ with 2 mA bias current from 1.8-V supply,

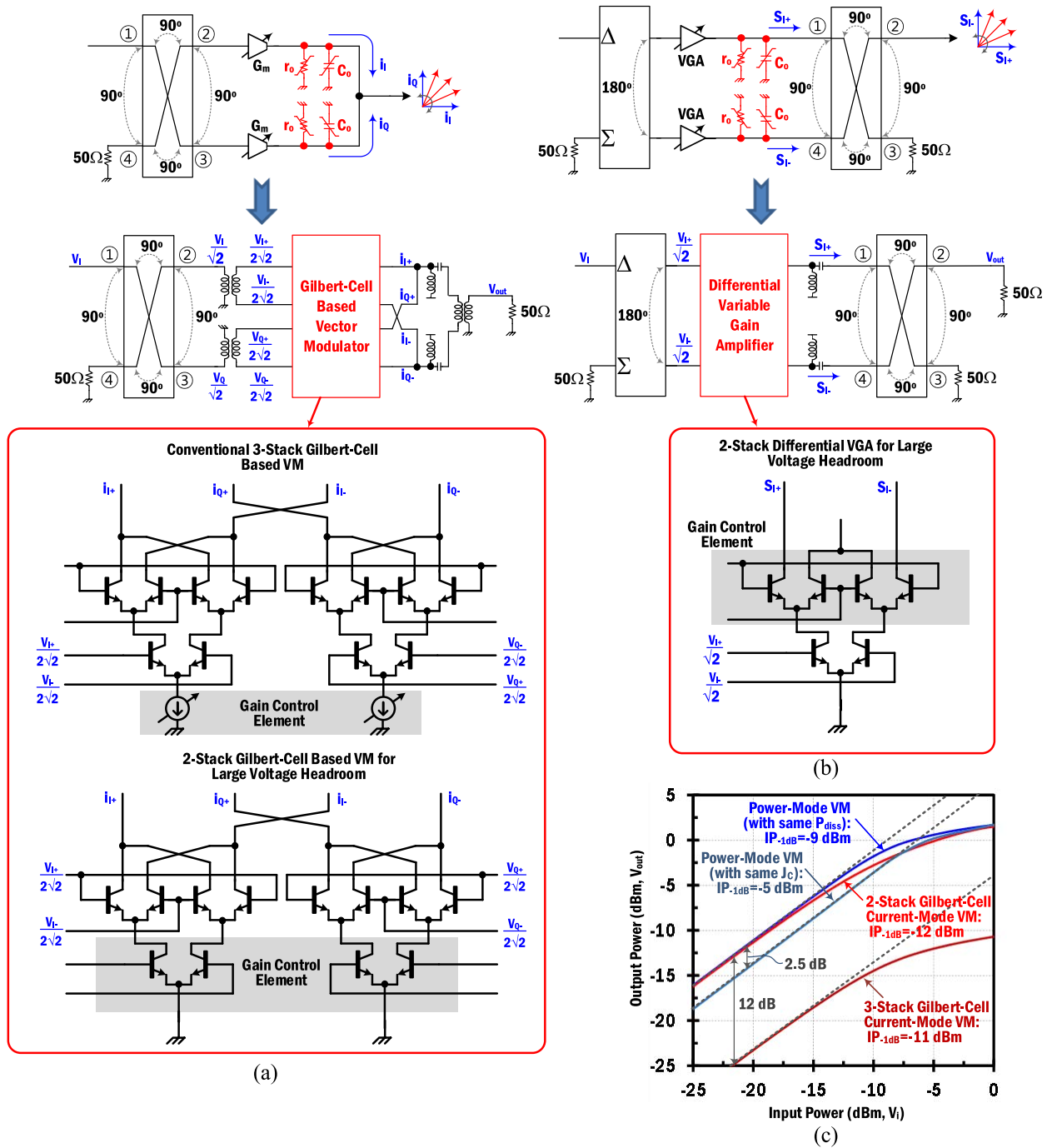


Fig. 4. Comparison of linearity performance. (a) Conventional Gilbert cell-based current-mode VM where the variable gain G_m cells outputs are loaded by each other nonlinear outputs. (b) Proposed differential VGA-based power-mode VM. (c) Linearity simulation results at 94 GHz in 0.13- μ m SiGe BiCMOS (the case for 45° phase state for both current-mode and power-mode VMs). The power dissipation is 8.8 mW from 2.2-V supply voltage for the Gilbert cell-based VMs (2 mA per each tail differential pair).

resulting in 3.6-mW dc power consumption per stage. The power dissipation in the bias path is less than 5% of the main signal path P_{diss} .

The noise and gain circles in the Smith chart shown in Fig. 7 reveals a tradeoff between NF and achievable gain. The LNA is matched close to 50 Ω and achieves 5–5.5-dB NF with 10-dB gain per stage. The interstage inductor (93 pH) between Q_1 and Q_2 in Fig. 5(a) resonates parasitic capacitance between the emitter and the ground of Q_2 , thus improving gain

and NF performances. About 600 fF of the bypass capacitor is added immediately after the 66 pH of load inductor to minimize parasitic inductance and to make a solid ac ground at 94 GHz. All inductor and transistor interconnects are modeled using SONNET software [27]. Fig. 5(b) shows full SONNET layout of the first stage LNA. A custom-made high- Q metal-oxide-metal (MOM) capacitor, made by top four metal layers, is used for output matching. The VGA after the phasing circuit in Fig. 2(b), is designed with the same cascode topology as

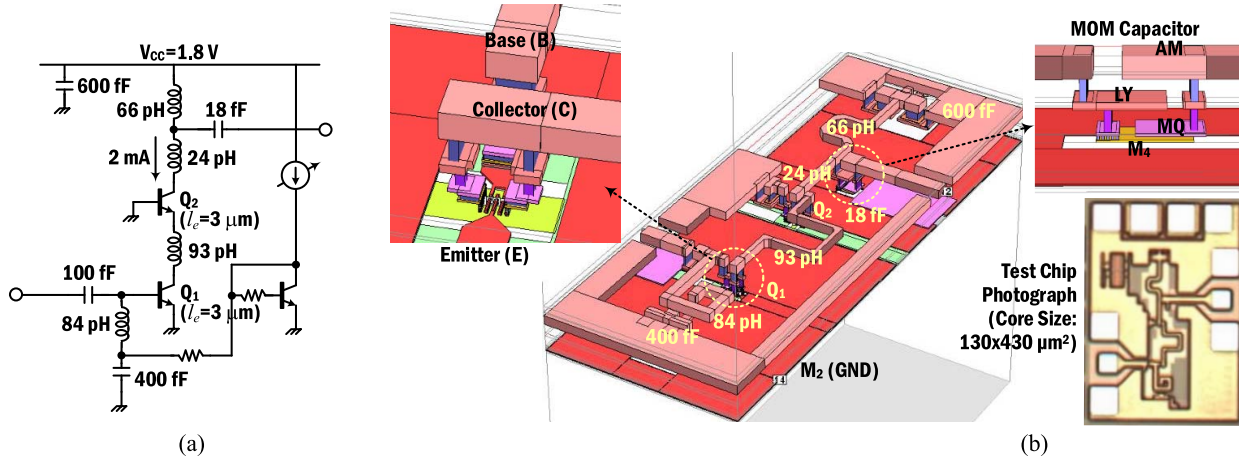


Fig. 5. (a) Schematic of the LNA and VGAs. (b) SONNET layout of the LNA showing transistor interconnection and output MOM capacitor for EM wave simulation and fabricated test chip photograph.

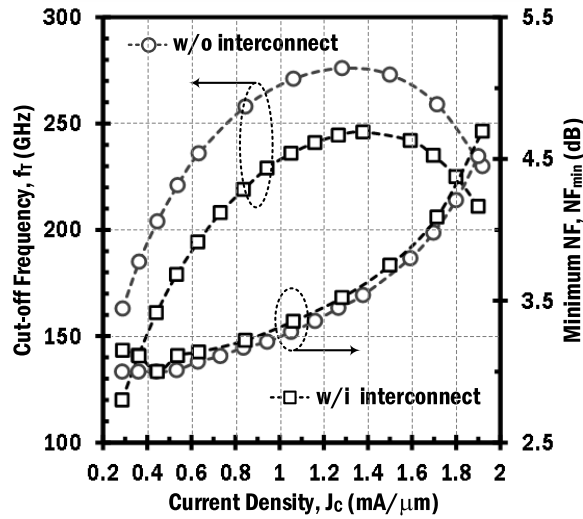


Fig. 6. Simulated cutoff frequency and minimum NF versus current density with and without transistor interconnects.

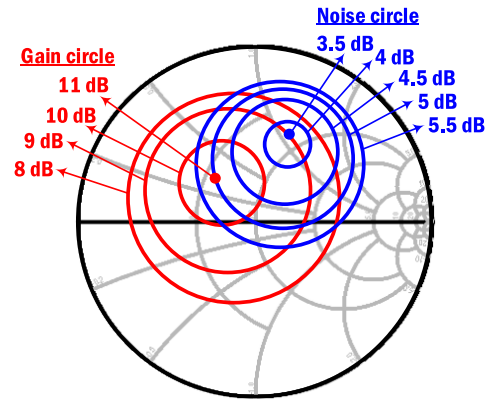


Fig. 7. Simulated noise and gain circles in the Smith chart at 94 GHz.

around 94 GHz, while there is incremental NF improvement as increasing J_c at 97 GHz. The measured input 1-dB gain compression point ($IP_{-1\text{dB}}$) is -15 dBm as shown in Fig. 8(d).

the LNA. The gain is variable by controlling the bias current of the amplifier and the gain variation is 5–10 dB with 1-dB gain step in simulations with a digital to analog converter (DAC) control, sufficient to correct the gain error of the PS.

The first stage of the LNA test chip is separately characterized [see Fig. 5(b)] and the measured performances are shown in Fig. 8. The measured input matching (S_{11}) is wideband due to a resistive 50Ω matching relying on the base input resistance of Q_1 . The measured impedance matching (S_{11} and S_{22}) and isolation (S_{12}) characteristics are well following the simulation trends in Fig. 8(a), although the matched frequency band of S_{22} is shifted to $\sim 97 \text{ GHz}$. The measured peak gain (S_{21}) is 9.7 dB and shifted to 97 GHz because of the output matching band shift. The measured gain at 94 GHz is 9 dB and 3-dB bandwidth (BW) is 90–107 GHz. The measured NF is 4.86–5.2 dB at 92–100 GHz [see Fig. 8(b)]. The NF dependence on J_c is characterized in Fig. 8(c), revealing that $J_c \sim 0.7 \text{ mA}/\mu\text{m}$ is near optimum

B. Phase Shifting Subsystem

1) $0^\circ/90^\circ$ Switchable LC Phase Shifter: The $0/90^\circ$ LC PS shown in Fig. 9(a) is based on a switchable low-pass network [23]. In the bypass state (0°), M_1 ($W/L = 24/0.13$) is ON and M_2 ($W/L = 48/0.13$) is OFF. The OFF state parasitic capacitance is resonated out by the inductor L_2 (42 pH). In the 90° delay state, M_1 is OFF while M_2 is ON. The size of the transistors is optimized to have a low loss. In the fabricated test pattern in Fig. 9(b), the core area is $90 \times 170 \mu\text{m}^2$. There is a good agreement between measurements and simulations that include full EM modeling. The measured quadrature phase error is less than 5° at 90–95 GHz and the measured loss of the PS ranges 3–4 dB over the 75–100 GHz. The simulated $IP_{-1\text{dB}}$ is 10 dBm.

2) 180° and 90° Hybrid Couplers: The 180° hybrid coupler is designed with lumped elements, as shown in Fig. 10(a), to reduce the area and the core occupies $245 \times 345 \mu\text{m}^2$ [see Fig. 10(b)]. In the 180° hybrid, the low-pass branch

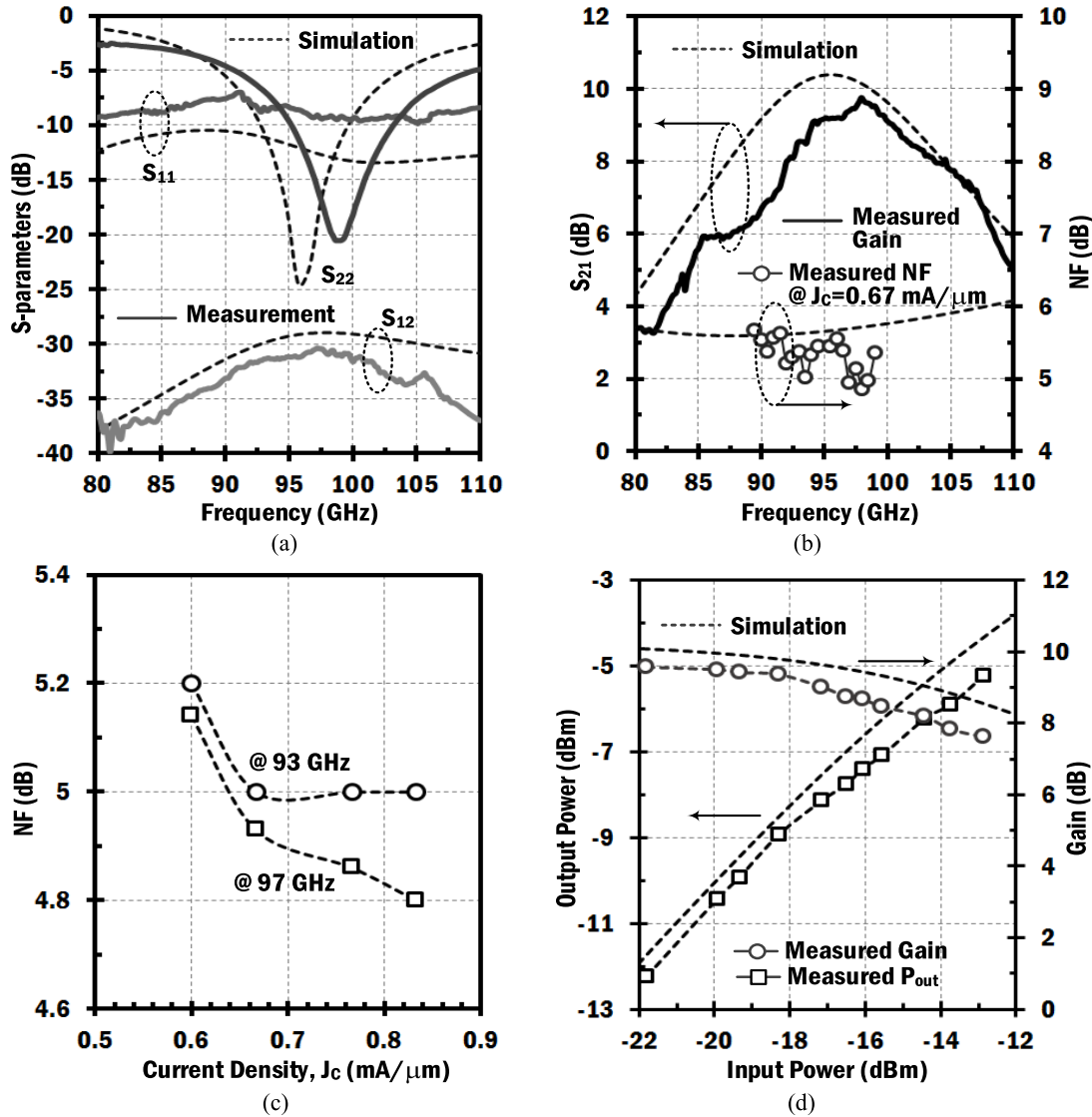


Fig. 8. Measurement results of the first stage LNA test circuit. (a) Input–output matching and isolation. (b) Gain and NF. (c) NF versus current density. (d) Linearity performance at 94 GHz.

(① $\Rightarrow$ ③) delays the signal phase, whereas the high-pass route (① $\Rightarrow$ ②) leads the signal phase, creating a phase difference between the two output nodes. By setting L_3 (120 pH) and C_2 (24 fF) to satisfy the impedance relationship of $\omega L_3 = 1/\omega C_2 = \sqrt{2} Z_o$ [28], the phase difference can be 180° with all four ports to be matched to 50Ω at 94 GHz. The S-parameter measurements were conducted on two test chips for measuring port ② (not shown) and port ③ response [see Fig. 10(b)]. The measurement results in Fig. 10(c) and (d) show that the loss is 0.7 dB at 94 GHz with less than 8° phase error over 90–110 GHz. The input/output matching (S_{11} and S_{22}) is < -10 dB for a wide frequency range of 85–105 GHz. The 90° hybrid coupler is based on meandered transmission lines [26] and causes about 1-dB loss with phase error $< 5^\circ$ over 84–118 GHz in the EM simulations.

3) *Quad Weighting VGAs*: The schematic of quad weighting amplifiers shown in Fig. 11(a) is compatible with layout floor plan, as can be seen in the test chip photograph. Each amplifier

has the same topology as LNA. The variable gain operation is achieved by steering dc current (I_{w0}) controlled by a DAC and the VGA can be turned ON and OFF by cutting off the bias current (I_{sw0}) using a switch network. The maximum current of each amplifier is 2 mA from 1.8-V supply for Rx channel and 2.5 V for Tx channel, resulting in the dc power consumption of 3.6 and 5 mW for the Rx and the Tx, respectively. Since two amplifiers are ON at a time, total dc power dissipation in the quad VGAs is 7.2 and 10 mW for the Rx and the Tx, respectively.

Fig. 11(b) shows the input and the output matching networks in detail. The OFF state impedance ($Z_{in,OFF}$) caused by parasitics of the OFF transistor is in parallel with the ON-state input impedance ($Z_{in,ON}$), resulting in a low input impedance of $Z_{in} = 12 - j17 \Omega$. In order to match the input to 50Ω , an inductor ($L_{IM} = 50 \text{ pH}$) is paralleled to keep the input reflection coefficient better than -10 dB. The asymmetry in the input network is included in the EM simulation when optimizing the input matching inductance—the asymmetry in

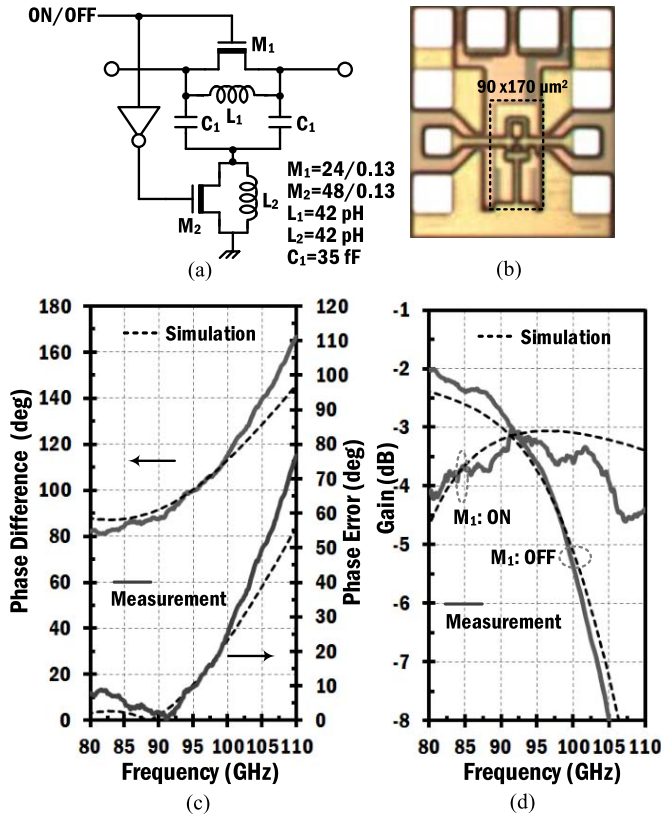


Fig. 9. Switchable $0^\circ/90^\circ$ LC PS. (a) Schematic design. (b) Fabricated chip photograph. (c) Measured phase difference and phase error from 90° . (d) Measured gain responses of the ON (0°) and OFF (90°) states of M_1 .

the input network does not affect the performance of the PS as long as the input reflection coefficient is below -10 dB.

While designing output matching network, about 15 ± 5 fF of C_p and 20 ± 5 pH of $L_{p,1-2}$ of parasitics are typical due to layout interconnects around the output of cascode transistors in Fig. 11(a). These results in a low capacitive impedance locus, $Z_{\text{out}} \Rightarrow \textcircled{x} \Rightarrow \textcircled{y} \Rightarrow \textcircled{z}$ in the Smith chart at the VGA output junction node [see Fig. 11(b)]. This low- Z parasitic network absorbs any transistor impedance disturbances caused by the VGA gain control, resulting in nearly constant $50\text{-}\Omega$ matching resistance over the VGA gain variations. In the test chip, the gain of the quad VGA is controlled by a 3-bit DAC—in actual T/R channel designs and a 4-bit DAC is used for fine gain control. The measurement result of the quad VGA for Rx channel is shown in Fig. 12. The measured gain variation ranges $-15 \sim 7$ dB, where the peak gain response is well matched with simulation [see Fig. 12(a)]. The measured S_{11} and S_{22} are < -10 dB at 92–100 GHz and almost constant for all the different gain states, resulting in negligible impedance variation at the input of the 90° hybrid phase interpolator.

As seen in Fig. 2, the Tx and Rx channels are based on the same phasing network except for supply voltage. However, the Tx PS is further optimized to reduce gain by decreasing Q of the interstage inductor L_M by inserting a small resistance ($\sim 10 \Omega$) in series with L_M . Also, the Rx channel supply voltage is 1.8 V but the Tx channel supply voltage is increased

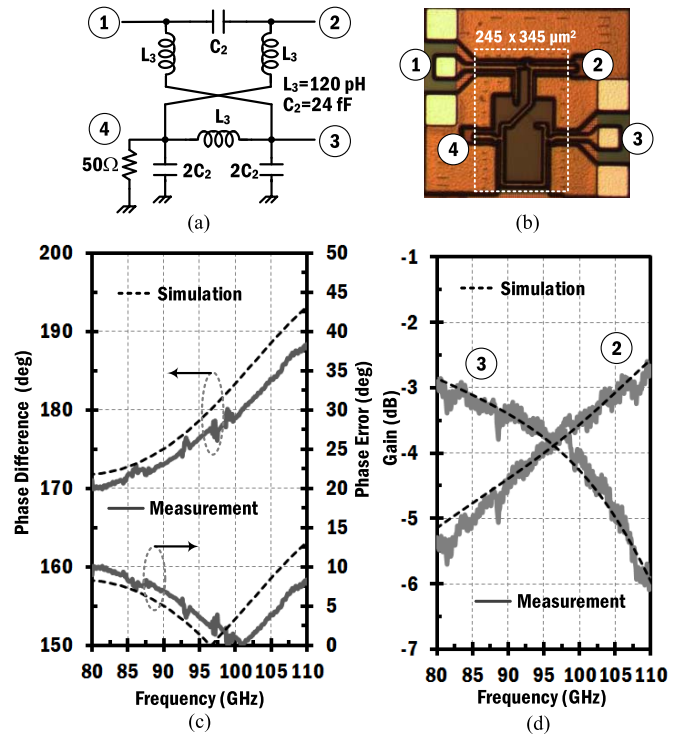


Fig. 10. Lumped 180° hybrid coupler. (a) Schematic design. (b) One of fabricated test chips photograph to characterize signal at port ③. (c) Measured phase difference and phase error from 180° between ports ② and ③. (d) Measured gain responses at ports ② and ③.

to 2.5 V to have better linearity. The PS linearity is constraint mainly by the quad VGAs and the switched LC PS has a negligible impact on this. The simulated Rx PS loss varies $0.5\text{--}5.5$ dB with maximum -3.9 dBm of $\text{IP}_{-1\text{dB}}$. For the Tx PS, the loss is higher and ranges $5\text{--}9$ dB. This loss is compensated by the following VGA. The simulated Tx PS $\text{IP}_{-1\text{dB}}$ is $4\text{--}6$ dBm.

C. Class-AB Power Amplifier

The power amplifier shown in Fig. 13(a) is a cascade of two class-AB common-emitter amplifiers. For a given large array applications, the required saturated output power (P_{sat}) per Tx phased-array element is ~ 5 dBm after SPDT. Thus, the required P_{sat} at the power amplifier (PA) output should be around 7 dBm when considering typical $1.5\text{--}2$ dB of the SPDT loss. One main barrier to maintain a high PAE in millimeter-wave PAs is the loss from passive components at the output stage which tends to increase as increasing frequency, especially in silicon process [29]–[44]. Therefore, it is important to set the optimum resistance (R_{opt}) to 50Ω after a dc feed inductor and dc block capacitor at the output stage to avoid any extra loss from the passive output impedance matching network. Once the supply voltage is set to a maximum which is limited by a transistor breakdown voltage (e.g., BV_{CEO} or BV_{CBO}), the output stage collector current will be determined by the required output power (e.g., $\text{OP}_{-1\text{dB}}$ or P_{sat}). The relatively small P_{sat} requires a low collector current, facilitating the setting R_{opt} to 50Ω . In the output stage, the HBT Q_2 is biased at a class-AB point with

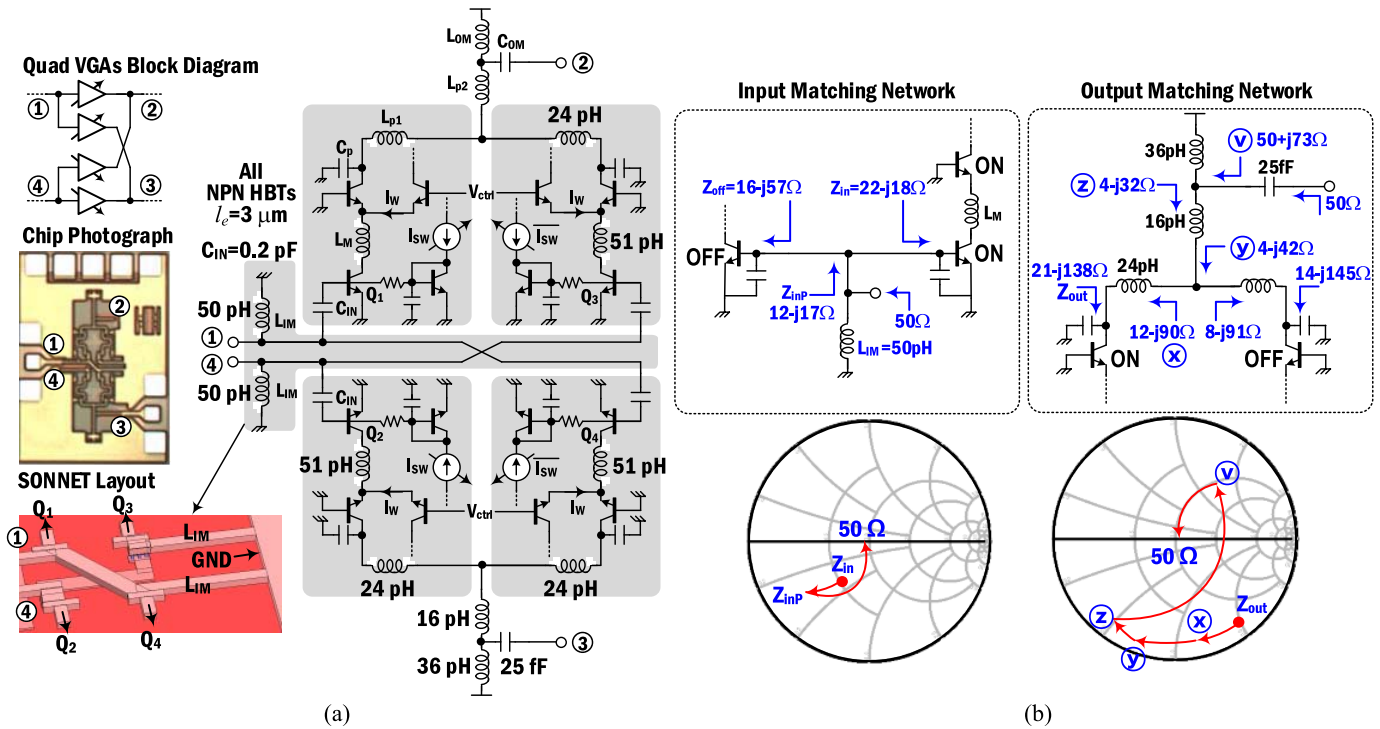


Fig. 11. Quad VGAs. (a) Block diagram, details on circuit schematic and input interconnect SONNET layout for EM simulation, and test chip photograph. (b) Input and output impedance matching details (all inductors are T-line-based inductor).

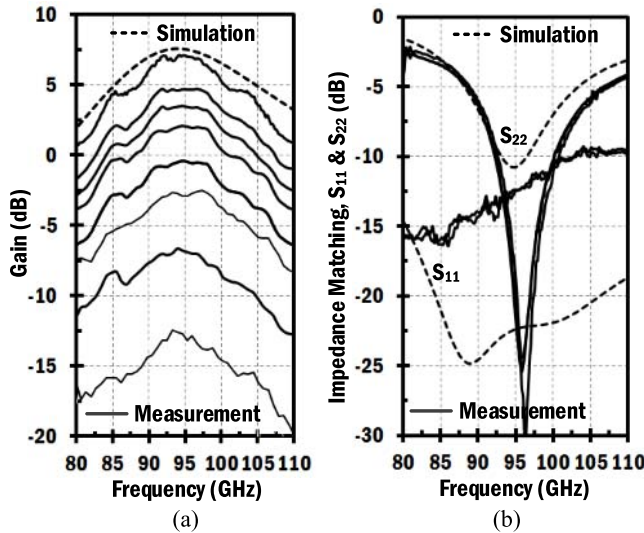


Fig. 12. Measured results of the quad VGA test chip. (a) Typical gain control. (b) Input and output impedance matching over all different gain states.

the bias current density of $J_C = 0.47 \text{ mA}/\mu\text{m}$ for high PAE, instead of a class-A bias point with $J_C = 1.3\text{--}1.4 \text{ mA}/\mu\text{m}$ for maximum cutoff frequency (f_T). Then, the emitter length is chosen to $9 \mu\text{m}$ to satisfy $\sim 7 \text{ dBm}$ P_{sat} .

When configuring the output matching circuit, the optimum load right after Q_2 , $Z_{\text{opt}1}$ in Fig. 13(b), is inductive due to parasitic capacitance at the collector node of Q_2 . Therefore, using a dc feed inductor (73 pH , $Q = 14$) the optimum load

can be located on the $50\text{-}\Omega$ unit circle, $Z_{\text{opt}2}$ in Fig. 13(b). Finally, cascading a dc block capacitor (55 fF , $Q = 17$) moves the optimum resistance to 50Ω . Note that the output matching circuit is in its simplest form by leveraging dc feed L and dc block C , resulting in a minimum loss from the passive network. In simulations, the P_{out} at a peak PAE point is 6.3 dBm with $\sim 2.3 \text{ dB}$ of saturated power gain in the output stage. As can be seen in Fig. 13(b), the PAE at the $50\text{-}\Omega$ matching point is better than 14% in the output stage—in a standalone output stage simulation, the peak PAE is 14.5%. The driver is designed to produce $\sim 4\text{-dBm}$ P_{out} with a highest possible PAE, $\sim 17.7\%$ PAE in Fig. 13(c). For this, the driver output is also matched for a maximum PAE, not for a maximum power transfer. In Fig. 13(c), first the impedance at the base of Q_2 , $Z_{\text{in}21}$ ($14.7 - j1.3 \Omega$), is moved to $Z_{\text{in}22}$ ($99 + j60 \Omega$) by a series T -line. Then, a series capacitor (42 fF) is cascaded to locate the impedance $Z_{\text{in}22}$ to $Z_{\text{in}23}$ ($132 - j16 \Omega$) close to the optimum impedance for a maximum PAE of Q_1 . Finally, a series capacitor (50 fF , $Q = 12$) followed by a shunt inductor (58 pH , $Q = 15$) matches the Q_1 input to 50Ω .

The cascaded performance of the two-stage PA is analyzed in Fig. 14. In Fig. 14(a), the first stage is loaded by the input impedance of the second stage, $Z_L = 122 - j47 \Omega$. The source impedance of the second stage is set to the conjugate of Z_L in Fig. 14(b) to deliver the output power of the first stage ($P_{\text{out}1}$) to the second stage without any loss. Based on this separation, the performance of each stage will be identical to that in the cascade PA. The simulated performance of each stage and cascaded PA at the peak PAE power at 94 GHz is

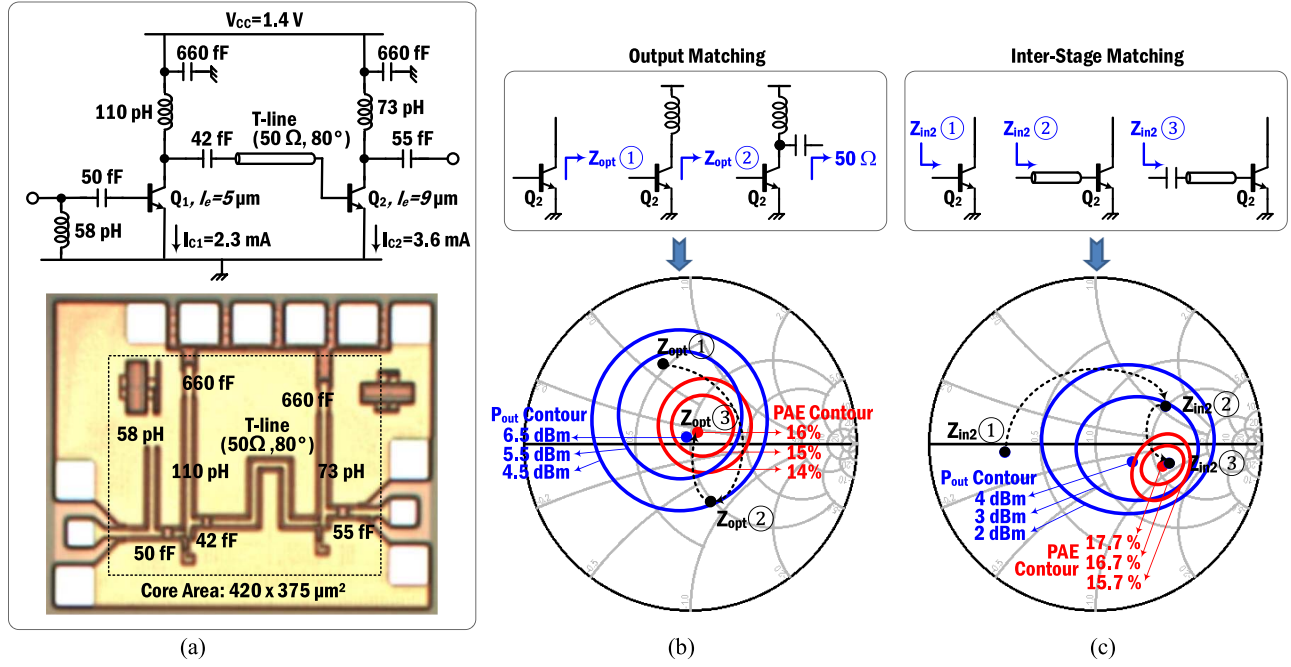


Fig. 13. Class-AB PA design. (a) PA schematic and fabricated test chip photograph. (b) Output matching network. (c) Interstage matching network.

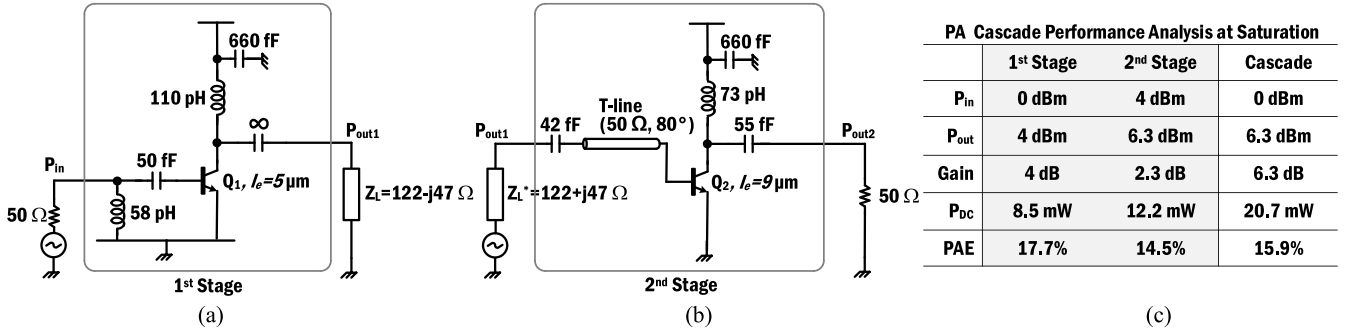


Fig. 14. Class-AB power amplifier cascade performance analysis at 94 GHz. (a) First stage input and output conditions. (b) Second stage input and output condition. (c) Performance analysis summary.

summarized in Fig. 14(c). In the cascade PA, the overall PAE is determined by

$$PAE = PAE_2 \times \left(1 + \frac{1}{G_{P1}} \frac{G_{P1} - 1}{G_{P2} - 1} \right) \times \left(1 + \frac{PAE_2}{PAE_1} \frac{1}{G_{P1}} \frac{G_{P1} - 1}{G_{P2} - 1} \right)^{-1} \quad (1)$$

where $G_{P1,2}$ are the power gains and $PAE_{1,2}$ are the PAE of the first and the second stage, respectively. The simulated PAE_1 is 17.7% and PAE_2 is 14.5%, resulting in 15.9% PAE from (1). This is well matched with ~16% of simulation PAE of the two-stage class-AB PA.

The PA is fully EM modeled using SONNET and characterized with a standalone test chip shown in Fig. 13(a). The small- and large-signal measurement results of the PA are shown in Fig. 15. The small-signal power gain (S_{21}) is 6.8 dB at 94 GHz with a 3-dB of 78.5–107.2 GHz. The measured impedance matching characteristics are $S_{11} < -10$ dB at

87–110 GHz and $S_{22} < -10$ dB at 80.3–108.3 GHz with the quiescent collector currents of $I_{C1}/I_{C2} = 2.3/3.6$ mA from 1.4-V supply voltage [see Fig. 15(a)]. In the large-signal characterizations, the measured P_{sat} is 7.4 dBm and OP_{-1dB} is 3.9 dBm at 94 GHz, as shown in Fig. 15(b). In the test chip, it is observed that the gain compression behavior is a bit milder than simulation and the PA achieves peak PAE of 18.2% at P_{sat} , a few percent better than the simulation. At this power level, the dc collector currents (I_{C1}/I_{C2}) are increased up to 5.5/9.6 mA, resulting in total 21.1 mW of dc power consumption in the two-stage PA. Finally, the PAE performance is compared with recent silicon PAs at similar frequency range in Fig. 15(c). It is apparent that the PAE decreases as decreasing output power due to a finite knee voltage effect, posing a challenge to achieve a high PAE at lower P_{out} . By simplifying the output matching network by setting 50-Ω R_{opt} and optimal interstage matching for a high PAE, the presented PA achieves one of the best PAEs even with a low P_{sat} .

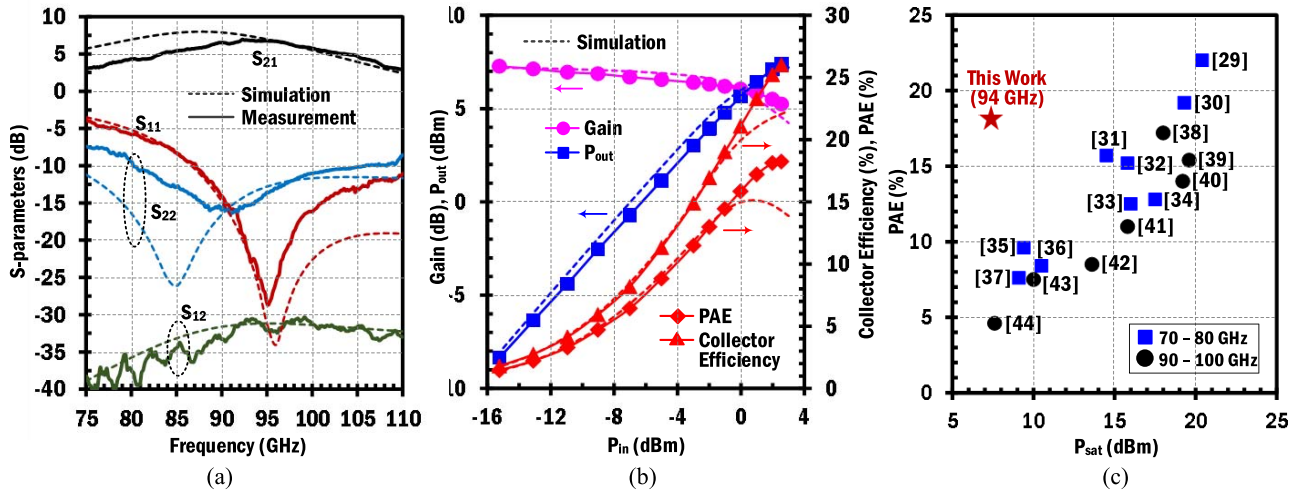


Fig. 15. Measurement results of the PA test chip. (a) Small-signal S-parameters. (b) Large-signal characteristics at 94 GHz. (c) PAE performance comparison.

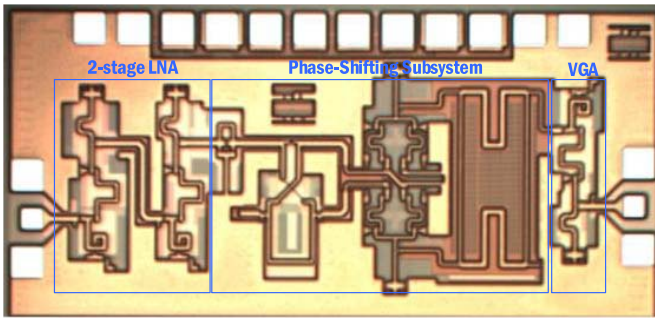


Fig. 16. Fabricated Rx channel chip photograph (chip size: $1.2 \times 0.55 \text{ mm}^2$ excluding I/O pads).

IV. EXPERIMENTAL RESULTS

The S-parameters of the T/R channels are measured after standard short-open-load-through (SOLT) calibration with W-band frequency extension modules. About 1 dB of the input and output ground-signal-ground (GSG) pad transition losses are de-embedded from the measurement data.

A. Receiver Channel

Fig. 16 shows fabricated Rx channel chip photograph and the core area of the Rx channel is $1.2 \times 0.55 \text{ mm}^2$. In the Rx channel measurement results, the input (S_{11}) and output (S_{22}) impedance matching characteristics shown in Fig. 17(a) are similar to those of the first stage LNA circuit in Fig. 8(a). This is because the LNA and the VGA in the Rx channel are based on the same topology. The measured peak gain ranges 17–23 dB over 4-bit phase states with the average gain of 21 dB and 3-dB BW of 92–100 GHz. As seen in Fig. 17(b), about 4.5 dB of gain drop from the simulation is observed at the target band. The single stage LNA gain discrepancy between simulation and measurement is ~ 1.5 dB [see Fig. 8(b)]. Since, the second stage of LNA and Rx VGA is based on the same design as the first stage LNA, we expect total 4.5 dB (1.5 dB per each amplifier stage) gain discrepancy in the overall Rx channel, well agreed with the test result

in Fig. 17(b). The rms gain error is < 1.4 dB at 90–100 GHz. The measured 4-bit phase response in Fig. 17(c) gives 2.4° rms phase error at 94 GHz and lower than 6° at 91–98 GHz.

The NF is measured by using a W-band noise source and the output signal is amplified by a W-band amplifier and then down-converted to an IF frequency using a W-band mixer. The IF signal is then measured by a noise figure meter after being amplified by an IF LNA and being filtered by a band pass filter (BPF). The noise calibration is performed including the cascaded network of W-band waveguides + W-band amplifier + mixer + IF LNA + IF BPF. The measured NF ranges 5.7–6.7 dB at 92–98 GHz and the NF variation over the 4-bit phase states is ~ 0.5 dB, as shown in Fig. 17(d). The gain drop does not affect the NF performance because the two-stage LNA gain is enough to reduce the noise effect from the following blocks. The typical measured $\text{IP}_{-1\text{dB}}$ with 0° phase setting is -31 dBm at 94 GHz (see Fig. 18). The total dc power consumption of the Rx channel is 18 mW from 1.8-V supply voltage.

B. Transmitter Channel

In the fabricated Tx channel shown in Fig. 19, the chip size is grown by 0.2 mm horizontally, resulting in $1.4 \times 0.55 \text{ mm}^2$, because of larger footprint of the two-stage class-AB PA than the two-stage LNA in the Rx channel. Note that the overall combined size of the Rx and the Tx channels is small enough to be integrated into the $\lambda/2 \times \lambda/2$ ($1.6 \times 1.6 \text{ mm}^2$) antenna array lattice with a room for SPDTs and other array interfaces at 94 GHz. The measured S_{11} and S_{22} are better than -8 dB over 90–110 GHz. The variation of S_{11} in Fig. 20(a) reflects the impedance variation at the input of the switched LC PS between the bypass and delay states. The measured gain peak is shifted to 96 GHz and the peak gain ranges 7.5–12 dB over 4-bit phase states with the average gain of 8.5 dB [see Fig. 20(b)]. At 94 GHz, the peak gain is 10 dB and the measured 3-dB BW is 92–100 GHz. The rms gain error is < 1.3 dB at 85–105 GHz. From the measured 4-bit phase response in Fig. 20(c), the rms phase error is 1.5° at 94 GHz and remains below 6° at 88–98 GHz.

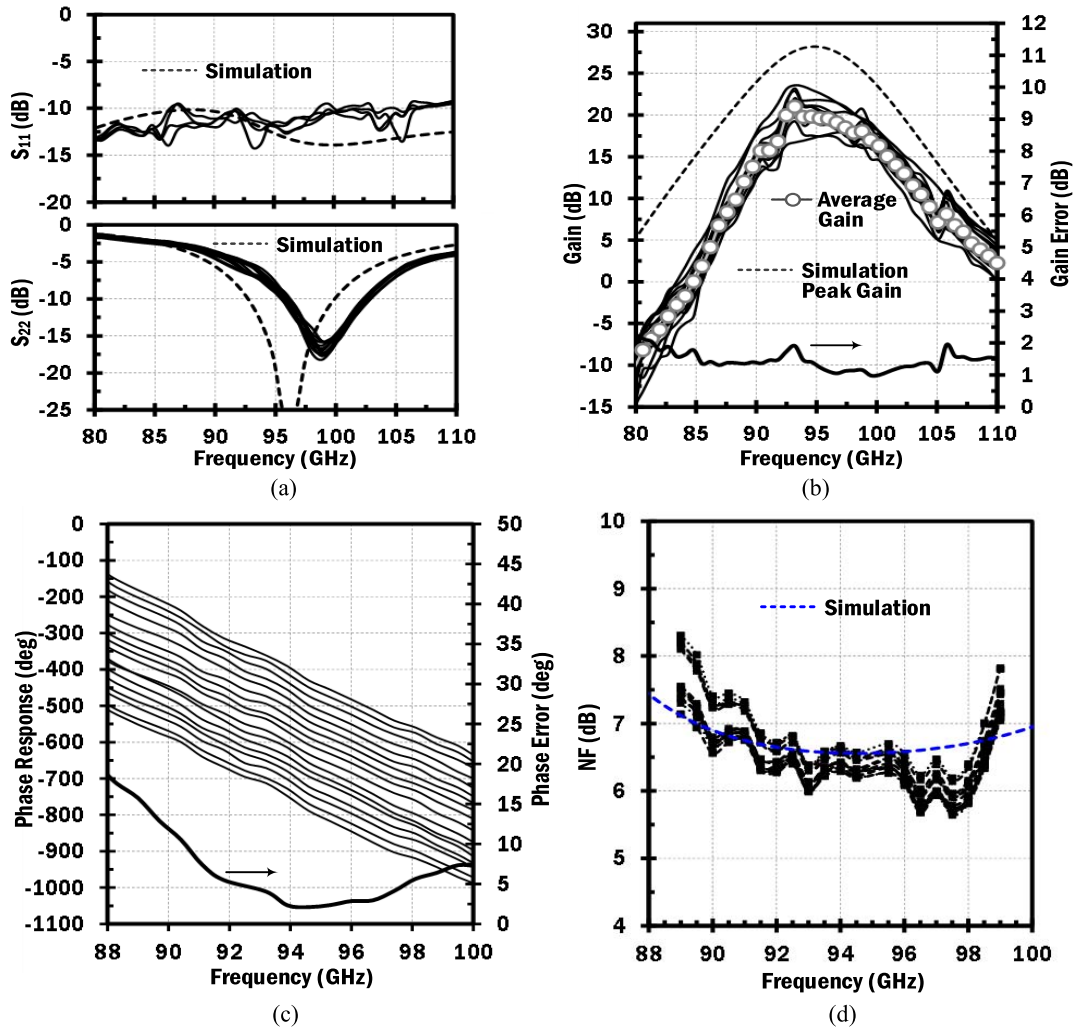


Fig. 17. Measured Rx channel characteristics. (a) Input (S_{11}) and output (S_{22}) matching. (b) Gain (S_{21}) responses for all 4-bit states. (c) 4-bit phase responses and rms phase error. (d) NF for all 4-bit phase states.

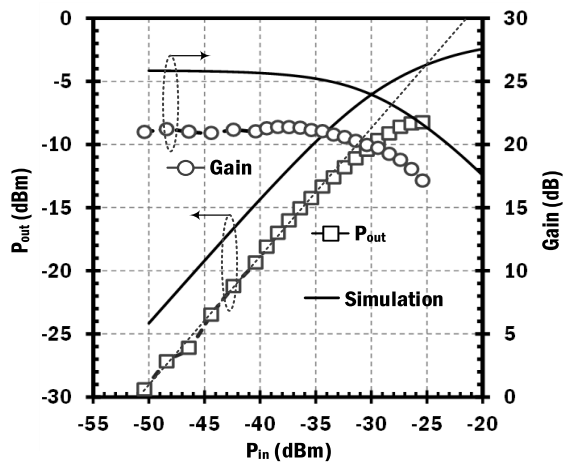


Fig. 18. Measured Rx channel linearity with 0° phase setting at 94 GHz.

The large-signal characteristics of the Tx channel shown in Fig. 20(d) is measured carefully at the highest gain state at 94 GHz. The linearity measurement is done using Agilent

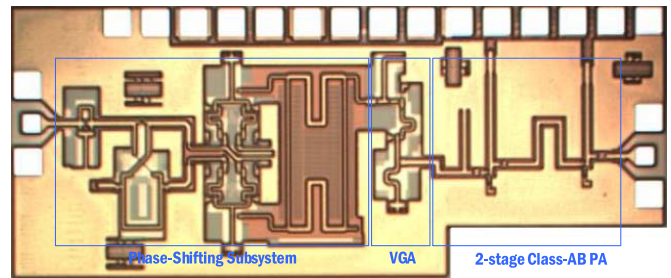


Fig. 19. Fabricated Tx channel chip photograph (chip size: $1.4 \times 0.55 \text{ mm}^2$ excluding I/O pads).

W-band power sensor and dual-channel power meter. All the losses at the input and the output of the Tx channel, including waveguide sections, attenuator, and probe loss, have been calibrated out. The measured P_{sat} is 7.5 dBm which is close to that of standalone PA in Fig. 15. However, the measured OP_{-1dB} decreases to 2.6 dBm ($IP_{-1dB} = -8.4 \text{ dBm}$), compared with 3.9 dBm of the PA's OP_{-1dB} , mainly because of nonlinearities of the preceding VGA and phasing circuits. The overall Tx

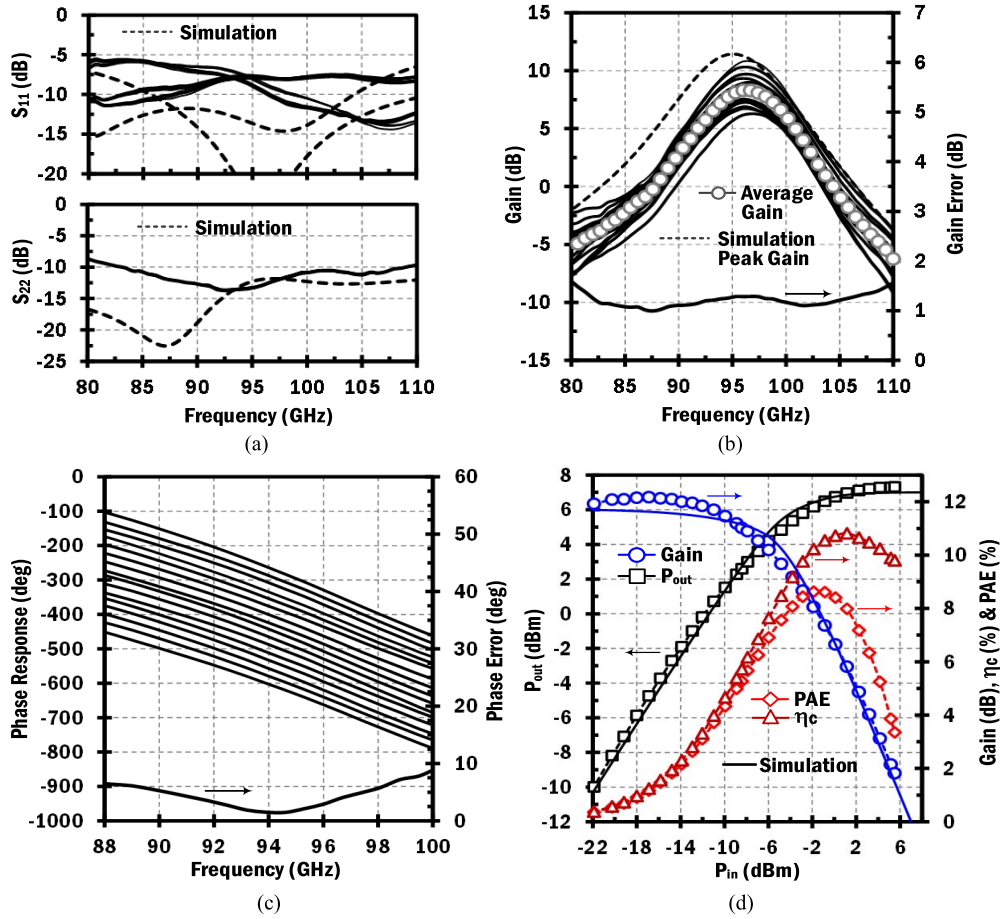


Fig. 20. Measured Tx channel characteristics. (a) Input (S_{11}) and output (S_{22}) matching. (b) Gain (S_{21}) responses for all 4-bit states. (c) 4-bit phase responses and rms phase error. (d) Large-signal characteristics at the highest gain state at 94 GHz.

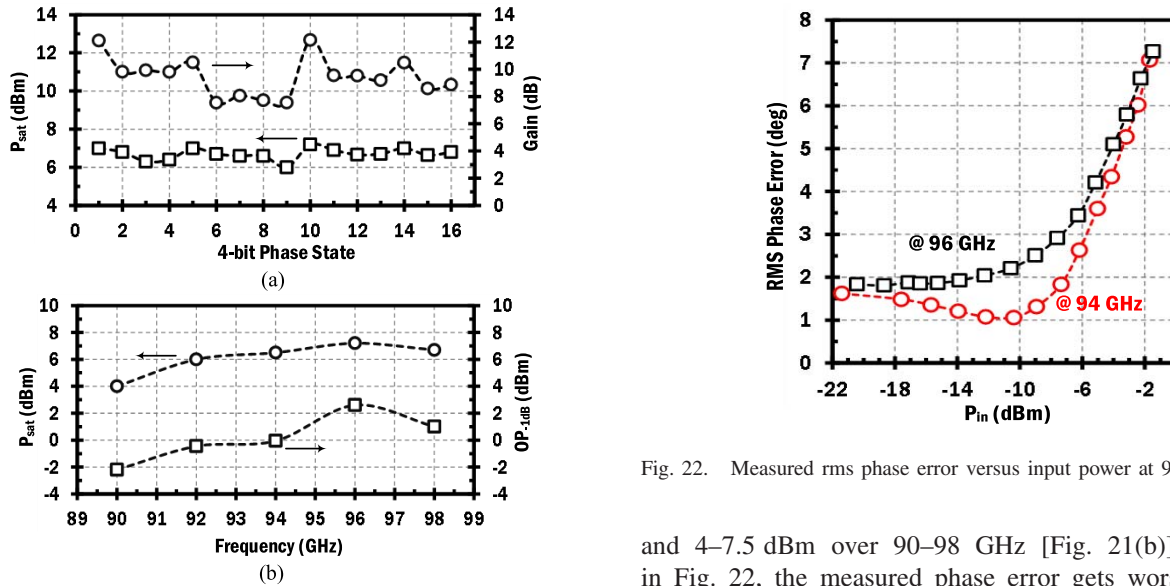


Fig. 21. Measured Tx channel gain, P_{sat} , and OP_{1dB} . (a) P_{sat} and gain over 4-bit phase states at 96 GHz. (b) P_{sat} , and OP_{1dB} versus frequency at the highest gain state at each frequency.

channel's collector efficiency (η_c) and PAE are 11% and 8%, respectively. The measured P_{sat} ranges 6.2–7.5 dBm over all the 4-bit phase states at 96 GHz [see Fig. 21(a)],

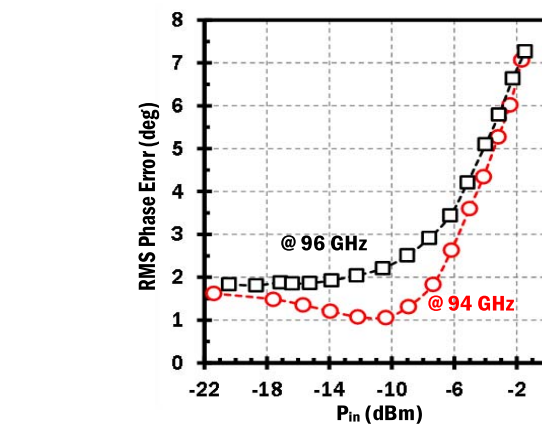


Fig. 22. Measured rms phase error versus input power at 94 and 96 GHz.

and 4–7.5 dBm over 90–98 GHz [Fig. 21(b)]. As shown in Fig. 22, the measured phase error gets worse after gain becomes compressed because of the amplitude modulation to phase modulation (AM-PM) phase distortion in the Tx channel. However, the measured rms phase error is $<8^\circ$ until $P_{in} = -2$ dBm, guaranteeing the 4-bit phased accuracy up to well above the gain saturation point. The dc power consumption of the Tx channel is 26 mW at the nominal bias condition of the PA.

TABLE I
RECEIVE CHANNEL PERFORMANCE SUMMARY AND COMPARISON WITH SOA WORKS

Reference	Freq. (GHz)	PS Topology	Gain (dB)	NF (dB)	IP _{-1dB} (dBm)	DR (dB·Hz)	P _{diss} (mW) / Channel	Rx FoM	Process
This Work	92-98	Passive VM (Power Combining)	17 – 23	5.8 – 6.6	-31.4	136 – 136.8	18	140.5 – 147	0.13μm SiGe BiCMOS
[20]	58.25 – 63.75	Active VM (Current Combining)	10 – 14	6.7 – 7.2	-16	151	78	141 – 146	65nm CMOS
[16]	78 – 97	Active VM (Current Combining)	19	7	-33	134	250*	129	0.18 μ m SiGe BiCMOS
[17]	90 – 100	Active VM (Current Combining)	26 – 29	9 – 11	-26 – -31	135	69	144.6	0.13 μ m SiGe BiCMOS
[7]	44 – 66	Active VM @Baseband (Current Combining)	26*	5.5	-27	141.5	30	152.7	45nm SOI CMOS
[8]	94	LO Phase Shifting	N/A	< 9.5	-19	145.5	56	-	0.13 μ m SiGe BiCMOS
[23]	76 – 84	Passive Switched LC	10 – 19	10.5	-26	137.5	33	132 – 141	0.13 μ m SiGe BiCMOS
[21]	58 – 65	RTPS	18	6.8 – 7.3	-28**	139	57	139 – 139.6	0.13 μ m SiGe BiCMOS
[18]	88 – 96	RTPS	30	8	-37	129	80	140	0.13 μ m SiGe BiCMOS

*Including mixer; **IP_{-1dB} of LNA only.

TABLE II
TRANSMIT CHANNEL PERFORMANCE SUMMARY AND COMPARISON WITH SOA WORKS

Reference	Freq. (GHz)	PS Topology	Gain (dB)	P _{sat} (dBm)	OP _{-1dB} (dBm)	PA Class	PA's Efficiency (%)	*P _{diss} (mW) / Channel	Process
This Work	92 – 98	Passive VM (Power Combining)	7.5 – 12	6 – 7.2	1.5 – 2.6	Class AB	18 (PAE) 26 (η_c)	26	0.13μm SiGe BiCMOS
[20]	60	Active VM (Current Combining)	7.7	8.3	4	N/A	4.4 (PAE)	168	65nm CMOS
[16]	78 – 97	Active VM (Current Combining)	19	8.5	N/A	N/A	N/A	250**	0.18 μ m SiGe BiCMOS
[17]	90 – 100	Active VM (Current Combining)	17	-4.4	N/A	Class A	3 (PAE, sim)	137	0.13 μ m SiGe BiCMOS
[6]	58.75 – 65	Active VM @Baseband (Current Combining)	N/A	-1.5	N/A	Class E	20 (η_c)	27	65nm CMOS
[14]	57 – 66	Passive Switched LC	< 0.5	7	5	Class A	N/A	100	65nm CMOS
[24]	57 – 65	RTPS	30	12	9	N/A	9 (PAE)	164	0.13 μ m SiGe BiCMOS
[18]	88 – 96	RTPS	> 25	6.3	3.3	Class A	6.7 (PAE)	116	0.13 μ m SiGe BiCMOS
[8]	94	LO Phase Shifting	N/A	6.5	N/A	Class AB	15 (PAE, sim)	71	0.13 μ m SiGe BiCMOS

*DC power dissipation at quiescent bias point; **including mixer.

C. Performance Comparison

Tables I and II compare performances of the proposed Rx and Tx channels with prior SOA phased-array elements at millimeter-wave frequencies. For a fair comparison of the Rx channel performance, taking into account of the Rx channel gain and dynamic range (DR), and power dissipation, the Rx performance figure of merit (Rx FoM) in Table I is defined as

$$\begin{aligned}
 \text{Rx FoM} &\triangleq \text{Gain} \times \frac{\text{DR}}{P_{\text{diss}}[\text{mW}]} \\
 &= \text{Gain}[\text{dB}] + 174[\text{dBm}] - \text{NF}[\text{dB}] + \text{IP}_{-1\text{dB}}[\text{dBm}] \\
 &\quad - 10\log(P_{\text{diss}}[\text{mW}]). \quad (2)
 \end{aligned}$$

The proposed Rx channel achieves one of the highest Rx FoMs (=147) at W-band with the smallest P_{diss} mainly because of a low loss in the power domain phasing system, compared with other SOAs in Table I. For the Tx channel, the quiescent point power dissipation is one of the smallest in Table II. The Tx channel P_{diss} will increase when the linear power gain increases to 15–20 dB, for instance, for more practical W-band phased-array front-end applications. However, when the saturated Tx channel power is capped at 5–8 dBm range in practical large array applications (number of elements > several thousands), the extra P_{diss} for boosting gain by any gain amplifier before the Tx PS will be incremental because

it is mostly a small-signal amplification. Thus, overall Tx channel P_{diss} will likely be smaller than that of the prior works based on RTPS, active Gilbert cell-based VM, or LO phase shifting architecture at W-band in Table II [8], [16]–[18].

V. CONCLUSION

This paper presents W-band phased-array T/R channels employing a 90°-hybrid phase interpolator based phase shifting system. The T/R channels are realized in 0.13- μm SiGe BiCMOS process. The proposed PS architecture exhibits high power efficiency because of having a low loss that can be compensated with relatively small dc power penalty, compared to prior pure passive or active PSs. Utilizing the proposed phasing system, the presented W-band T/R channels achieve a record Rx FoM and the Tx channel efficiency at the expense of some of the lowest dc power consumptions, compared with the prior SOA T/R channels in silicon technology. The high power efficiency is particularly promising for relieving thermal management issues in large-scale silicon phased arrays at millimeter-wave frequencies.

ACKNOWLEDGMENT

The authors would like to thank H. A. Hung in the U.S. Army Research Laboratory for supporting the W-band test equipment.

REFERENCES

- [1] K. K. Tokgoz *et al.*, "13.3 A 56 Gb/s W-band CMOS wireless transceiver," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Jan./Feb. 2016, pp. 242–243.
- [2] F. Caster, L. Gilreath, S. Pan, Z. Wang, F. Capolino, and P. Heydari, "Design and analysis of a W-band 9-element imaging array receiver using spatial-overlapping super-pixels in silicon," *IEEE J. Solid-State Circuits*, vol. 49, no. 6, pp. 1317–1332, Jun. 2014.
- [3] H. B. Wallace, "An application of advanced SiGe to millimeter-wave phased arrays," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Montreal, QC, Canada, Jun. 2012, pp. 1–3.
- [4] S. Kishimoto, N. Orihashi, Y. Hamada, M. Ito, and K. Maruhashi, "A 60-GHz band CMOS phased array transmitter utilizing compact baseband phase shifters," in *Proc. IEEE Radio Freq. Integr. Circuits Symp.*, Jun. 2009, pp. 215–218.
- [5] M. Tabesh *et al.*, "A 65 nm CMOS 4-element sub-34 mW/element 60 GHz phased-array transceiver," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2011, pp. 166–168.
- [6] M. Tabesh *et al.*, "A 65 nm CMOS 4-element sub-34 mW/element 60 GHz phased-array transceiver," *IEEE J. Solid-State Circuits*, vol. 46, no. 12, pp. 3018–3032, Dec. 2011.
- [7] S. Kundu and J. Paramesh, "A compact, supply-voltage scalable 45–66 GHz baseband-combining CMOS phased-array receiver," *IEEE J. Solid-State Circuits*, vol. 50, no. 2, pp. 527–542, Feb. 2015.
- [8] A. Townley *et al.*, "A 94-GHz 4TX–4RX phased-array FMCW radar transceiver with antenna-in-package," *IEEE J. Solid-State Circuits*, vol. 52, no. 5, pp. 1245–1259, May 2017.
- [9] W. L. Chan, J. R. Long, M. Spirito, and J. J. Pekarik, "A 60 GHz-band 2 $\times$ 2 phased-array transmitter in 65 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2010, pp. 42–43.
- [10] H. Jia *et al.*, "A 77 GHz frequency doubling two-path phased-array FMCW transceiver for automotive radar," *IEEE J. Solid-State Circuits*, vol. 51, no. 10, pp. 2299–2311, Oct. 2016.
- [11] A. Natarajan, A. Komijani, X. Guan, A. Babakhani, and A. Hajimiri, "A 77-GHz phased-array transceiver with on-chip antennas in silicon: Transmitter and local lo-path phase shifting," *IEEE J. Solid-State Circuits*, vol. 41, no. 12, pp. 2807–2819, Dec. 2006.
- [12] S. Emami *et al.*, "A 60 GHz CMOS phased-array transceiver pair for multi-Gb/s wireless communications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2011, pp. 164–166.
- [13] E. Cohen, C. Jakobson, S. Ravid, and D. Ritter, "A bidirectional TX/RX four-element phased array at 60 GHz with RF-IF conversion block in 90-nm CMOS process," *IEEE Trans. Microw. Theory Techn.*, vol. 58, no. 5, pp. 1438–1446, May 2010.
- [14] J.-L. Kuo *et al.*, "60-GHz four-element phased-array transmit/receive system-in-package using phase compensation techniques in 65-nm flip-chip CMOS process," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 3, pp. 743–756, Mar. 2012.
- [15] B. H. Ku, O. Inac, M. Chang, H. H. Yang, and G. M. Rebeiz, "A high-linearity 76–85-GHz 16-element 8-transmit/8-receive phased-array chip with high isolation and flip-chip packaging," *IEEE Trans. Microw. Theory Techn.*, vol. 62, no. 10, pp. 2337–2356, Oct. 2014.
- [16] S. Shahramian, Y. Baeyens, N. Kaneda, and Y.-K. Chen, "A 70–100 GHz direct-conversion transmitter and receiver phased array chipset demonstrating 10 Gb/s wireless link," *IEEE J. Solid-State Circuits*, vol. 48, no. 5, pp. 1113–1125, May 2013.
- [17] F. Golcuk, T. Kanar, and G. M. Rebeiz, "A 90–100-GHz 4 $\times$ 4 SiGe BiCMOS polarimetric transmit/receive phased array with simultaneous receive-beams capabilities," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 8, pp. 3099–3114, Aug. 2013.
- [18] A. Natarajan, A. Valdes-Garcia, B. Sadhu, S. K. Reynolds, and B. D. Parker, "W-band dual-polarization phased-array transceiver front-end in SiGe BiCMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 63, no. 6, pp. 1989–2002, Jun. 2015.
- [19] W. Shin, B.-H. Ku, O. Inac, Y.-C. Ou, and G. M. Rebeiz, "A 108–114 GHz 4 $\times$ 4 wafer-scale phased array transmitter with high-efficiency on-chip antennas," *IEEE J. Solid-State Circuits*, vol. 48, no. 9, pp. 2041–2055, Sep. 2013.
- [20] Y. Yu, P. G. M. Baltus, A. De Graauw, E. van der Heijden, C. S. Vaucher, and A. H. M. van Roermund, "A 60 GHz phase shifter integrated with LNA and PA in 65 nm CMOS for phased array systems," *IEEE J. Solid-State Circuits*, vol. 45, no. 9, pp. 1697–1709, Sep. 2010.
- [21] A. Natarajan *et al.*, "A fully-integrated 16-element phased-array receiver in SiGe BiCMOS for 60-GHz communications," *IEEE J. Solid-State Circuits*, vol. 46, no. 5, pp. 1059–1075, May 2011.
- [22] S. Y. Kim, O. Inac, C.-Y. Kim, D. Shin, and G. M. Rebeiz, "A 76–84-GHz 16-element phased-array receiver with a chip-level built-in self-test system," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 8, pp. 3083–3098, Aug. 2013.
- [23] S. Y. Kim and G. M. Rebeiz, "A low-power BiCMOS 4-element phased array receiver for 76–84 GHz radars and communication systems," *IEEE J. Solid-State Circuits*, vol. 47, no. 2, pp. 359–367, Feb. 2012.
- [24] A. Valdes-Garcia *et al.*, "A fully integrated 16-element phased-array transmitter in SiGe BiCMOS for 60-GHz communications," *IEEE J. Solid-State Circuits*, vol. 45, no. 12, pp. 2757–2773, Dec. 2010.
- [25] S. Zahir, O. D. Gurbuz, A. Kar-Roy, S. Raman, and G. M. Rebeiz, "60-GHz 64- and 256-elements wafer-scale phased-array transmitters using full-reticle and subreticle stitching techniques," *IEEE Trans. Microw. Theory Techn.*, vol. 64, no. 12, pp. 4701–4719, Dec. 2016.
- [26] S. Afroz and K.-J. Koh, "90° hybrid-coupler based phase-interpolation phase-shifter for phased-array applications at W-band and beyond," in *IEEE MTT-S Int. Microw. Symp. Dig.*, San Francisco, CA, USA, May 2016, pp. 1–4.
- [27] *Sonnet, Versio 15.54*, Sonnet Softw. Inc., Syracuse, NY, USA, 2018.
- [28] S. J. Parisi, "180° lumped element hybrid," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 1989, pp. 1243–1246.
- [29] D. Zhao and P. Reynaert, "An E-band power amplifier with broadband parallel-series power combiner in 40-nm CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 63, no. 2, pp. 683–690, Feb. 2015.
- [30] K.-Y. Wang, T.-Y. Chang, and C.-K. Wang, "A 1 V 19.3 dBm 79 GHz power amplifier in 65 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2012, pp. 260–262.
- [31] S. T. Nicolson *et al.*, "A low-voltage SiGe BiCMOS 77-GHz automotive radar chipset," *IEEE Trans. Microw. Theory Techn.*, vol. 56, no. 5, pp. 1092–1104, May 2008.
- [32] J. Oh, B. Ku, and S. Hong, "A 77-GHz CMOS power amplifier with a parallel power combiner based on transmission-line transformer," *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 7, pp. 2662–2669, Jul. 2013.
- [33] R. Ben Yishay, R. Carmon, O. Katz, and D. Elad, "A high gain wideband 77GHz SiGe power amplifier," in *Proc. IEEE Radio Freq. Integr. Circuits Symp.*, Anaheim, CA, USA, May 2010, pp. 529–532.
- [34] A. Komijani and A. Hajimiri, "A wideband 77-GHz, 17.5-dBm fully integrated power amplifier in silicon," *IEEE J. Solid-State Circuits*, vol. 41, no. 8, pp. 1749–1756, Aug. 2006.

- [35] Y. Hamada, M. Tanomura, M. Ito, and K. Maruhashi, "A high gain 77 GHz power amplifier operating at 0.7 V based on 90 nm CMOS technology," *IEEE Microw. Wireless Compon. Lett.*, vol. 19, no. 5, pp. 329–331, May 2009.
- [36] Y.-A. Li, M.-H. Hung, S.-J. Huang, and J. Lee, "A fully integrated 77 GHz FMCW radar system in 65 nm CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2010, pp. 216–217.
- [37] J.-J. Lin, K.-H. To, D. Hammock, B. Knappenberger, M. Majerus, and W. M. Huang, "Power amplifier for 77-GHz automotive radar in 90-nm LP CMOS technology," *IEEE Microw. Wireless Compon. Lett.*, vol. 20, no. 5, pp. 292–294, May 2010.
- [38] P. Song, A. Ç. Ulusoy, R. L. Schmid, S. N. Zeinolabedinzadeh, and J. D. Cressler, "W-band SiGe power amplifiers," in *Proc. IEEE Bipolar/BiCMOS Circuits Technol. Meeting (BCTM)*, Coronado, CA, USA, Sep./Oct. 2014, pp. 151–154.
- [39] M. Chang and G. M. Rebeiz, "A wideband high-efficiency 79–97 GHz SiGe linear power amplifier with $\gg 90$ mW output," in *Proc. IEEE Bipolar/BiCMOS Circuits Technol. Meeting*, Monterey, CA, USA, Oct. 2008, pp. 69–72.
- [40] A. Agah, J. A. Jayamon, P. M. Asbeck, L. E. Larson, and J. F. Buckwalter, "Multi-drive stacked-FET power amplifiers at 90 GHz in 45 nm SOI CMOS," *IEEE J. Solid-State Circuits*, vol. 49, no. 5, pp. 1148–1157, May 2014.
- [41] A. Agah, J. Jayamon, P. Asbeck, J. Buckwalter, and L. Larson, "A 11% PAE, 15.8-dBm two-stage 90-GHz stacked-FET power amplifier in 45-nm SOI CMOS," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Seattle, WA, USA, Jun. 2013, pp. 1–3.
- [42] Q. J. Gu, Z. Xu, and M.-C. F. Chang, "Two-way current-combining W-band power amplifier in 65-nm CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 5, pp. 1365–1374, May 2012.
- [43] D. Sandstrom, M. Varonen, M. Karkkainen, and K. Halonen, "W-band CMOS amplifiers achieving +10 dBm saturated output power and 7.5 dB NF," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2009, pp. 486–487.
- [44] N. Deferm, J. F. Osorio, A. de Graauw, and P. Reynaert, "A 94 GHz differential power amplifier in 45 nm LP CMOS," in *Proc. IEEE Radio Freq. Integr. Circuits Symp.*, Baltimore, MD, USA, Jun. 2011, pp. 1–4.



Sadia Afroz (S'10) received the B.S. degree in electrical and electronics engineering from the Bangladesh University of Engineering and Technology, Dhaka, Bangladesh, in 2011, and the Ph.D. degree in electrical engineering from the Virginia Polytechnic Institute and State University, Blacksburg, VA, USA, in 2017.

She is currently a Senior RF/FE Design Engineer with Qualcomm, San Diego, CA, USA. Her current research interests include RF and millimeter-wave circuits, transceivers and phased-array systems, and

high-speed ultrawideband interchip and intrachip wireless communication.

Dr. Afroz was a recipient of the Best Student Poster Award in CESA/MICS day from the Electrical and Computer Engineering Department, Virginia Polytechnic Institute and State University, in 2017.



Hyunchul Kim (S'15) received the B.S. and M.S. degrees in electrical engineering from Korea University, Seoul, South Korea, in 2009 and 2013, respectively. He is currently pursuing the Ph.D. degree in electrical and computer engineering with the Virginia Polytechnic Institute and State University, Blacksburg, VA, USA.

His current research interests include millimeter-wave-integrated circuits and systems.



Kwang-Jin Koh (S'06–M'09) received the Ph.D. degree in electrical and computer engineering from the University of California at San Diego, La Jolla, CA, USA, in 2008.

From 2000 to 2004, he was a Research Staff with the Electronics and Telecommunications Research Institute, Daejeon, South Korea. From 2008 to 2010, he was a Senior Engineer with Intel Corporation, Hillsboro, OR, USA. From 2010 to 2011, he was a Senior Staff Scientist with Broadcom Corporation, Irvine, CA, USA. He is currently an Assistant Professor with the Electrical and Computer Engineering Department, Virginia Polytechnic Institute and State University, Blacksburg, VA, USA. His current research interests include analog and mixed-mode circuits, antennas, and applied electromagnetics from RF to millimeter-wave and terahertz range.

He and his doctoral advisees were a recipient of the First Place Best Paper Award in the Advanced Practice Paper Competition and the Honorable Mention Award in the Student Paper Competition in the 2017 IEEE/MTT-S International Microwave Symposium (IMS), the First Place Best Student Paper Award in the 2016 IEEE/MTT-S IMS, and the Second Place Best Student Paper Award in the 2015 IEEE/MTT-S IMS. He was a recipient of the Best Paper Award of the IEEE Solid-State Circuits and the IEEE Electron Device Societies, Seoul Chapter in 2002, Science, Technology, Engineering, and Mathematics Research Fellowships from the National Aeronautics and Space Administration and the Commonwealth of Virginia from 2015 to 2017, the Outstanding Assistant Professor Award from the College of Engineering in 2014, and the Junior Faculty Research Award from the Institute for Critical Technology and Applied Science in 2012, Virginia Polytechnic Institute and State University. He was a finalist of the R.W.P. King Award of the IEEE Antenna and Propagation Society in 2015. He was also one of the recipients of the Team of the Year Award in 2010 from the Teledyne Technology Inc., (formerly, Rockwell Scientific Corporation) in recognition of his microwave and millimeter-wave silicon phased arrays which were reported to the U.S. Department of Defense by the Defense Advanced Research Projects Agency as one of the agency's major achievements in 2007–2008.

Dr. Koh has served or has been serving as a Technical Program Committee Member of the IEEE Bipolar/BiCMOS Circuits and Technology Meeting, the IEEE Custom Integrated Circuits Conference, the IEEE/MTT-S IMS, and the IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium.